

RL Model Status

Legacy PPO Research vs Current Production

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1 Current Status

PPO/RL is not part of the active live trading stack. The active live stack is the JEPA event-option static-union package. The bot uses policy gates, static caps, cooldowns, and mechanical exits, not a PPO exit head.

2 Legacy RL Stack

The legacy RL implementation lives under:

```
neural/rl/  
backtest/backtest_rl.py  
neural/run_pipeline.ps1
```

It was designed to select option strikes and exits after a GBT entry signal. The architecture included a PPO agent with strike and exit heads plus a critic.

3 Why It Is Not Live

The RL layer historically showed failure modes:

- exit-head collapse;
- premature exits;
- worse drawdown than simpler baselines;
- difficulty matching live option-chain constraints;
- sensitivity to reward scaling and forced hold transitions.

The current production package avoids these degrees of freedom and uses an explicit mechanical exit:

```
stop=-60%  
tp=1000%  
trail=50%/25%  
min_hold=30m  
max_hold=180m
```

4 When To Revisit RL

RL can be reconsidered only as a candidate replacement or overlay. It must beat the static-union event-option package under the same gates:

- profit factor at least 1.3;
- win rate at least 45%;
- completed-month volume at least the production gate;
- no feature leakage;
- live-equivalent feature and option-chain construction;
- successful systemd startup and paper-intent replay.

5 Useful Legacy Lessons

The old RL work is still useful for:

- defining realistic option-chain environments;
- studying min-hold constraints;
- evaluating strike-selection headroom;
- comparing mechanical exits to learned exits;
- building future supervised or offline-RL candidate models.

6 Production Rule

Do not wire RL into `systemd/ai_bot.service` unless a new production package and validation report explicitly promote it. The July 2026 service must continue to load the event-option policy and component registry.